

When Does Aggregating Explanations Work?

OctMNIST Results

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This report consolidates the released `OctMNIST` experiment results. Tables are grouped by model, experimental protocol, and aggregation setting. Metric directions and robustness definitions are stated in every table header and caption.

Protocol	Model	Settings
Full-matrix protocol	ViT-B/16	NAIVE, IND, NOISE-S, NOISE-K

Table 1: NAIVE results for OctMNIST with ViT-B/16 (Full-matrix protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized. The test split contains $n = 1000$ samples.

Method	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
	F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	$R_F^g \uparrow$	$R_{\bar{F}}^g \uparrow$	$R_C^g \uparrow$	$R_{\bar{C}}^g \uparrow$	$R_F^p \uparrow$	$R_{\bar{F}}^p \uparrow$	$R_C^p \uparrow$	$R_{\bar{C}}^p \uparrow$	$R_F^s \uparrow$	$R_{\bar{F}}^s \uparrow$	$R_C^s \uparrow$	$R_{\bar{C}}^s \uparrow$	$R_F^a \uparrow$	$R_{\bar{F}}^a \uparrow$	$R_C^a \uparrow$	$R_{\bar{C}}^a \uparrow$
Best Individual	0.262	0.194	0.309	0.217	0.047	-0.044	0.077	-0.120	0.080	-0.007	0.079	-0.085	0.008	0.012	0.008	0.001	-0.025	-0.064	-0.021	-0.066
Simple Averaging	0.245	0.219	0.280	0.247	0.012	-0.065	0.052	-0.153	0.070	-0.013	0.087	-0.090	0.002	0.012	0.000	0.001	0.004	-0.129	-0.003	-0.138
Borda	0.226	0.261	0.261	0.298	0.019	-0.022	0.055	-0.099	0.060	0.007	0.066	-0.045	-0.005	0.007	-0.001	0.000	-0.010	-0.107	-0.016	-0.106
Kemeny–Young	0.217	0.240	0.261	0.263	0.034	-0.023	0.073	-0.100	0.056	0.013	0.063	-0.048	0.003	0.011	0.004	0.003	-0.024	-0.113	-0.032	-0.120
RRF	0.216	0.235	0.252	0.266	0.033	-0.037	0.074	-0.109	0.078	0.023	0.085	-0.042	0.008	0.002	0.012	-0.004	-0.002	-0.103	-0.006	-0.106
Schulze	0.216	0.251	0.256	0.277	0.037	-0.014	0.078	-0.091	0.072	0.035	0.076	-0.035	0.006	0.021	0.006	0.018	-0.009	-0.108	-0.007	-0.111

Table 2: IND results for OctMNIST with ViT-B/16 (Full-matrix protocol). The Best Individual row is followed by matched NAIVE and IND results for each aggregation method at $q = 11$. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized. The test split contains $n = 1000$ samples.

Method	Setting	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _F ^g $\uparrow$	R _C ^g $\uparrow$	R _C ^g $\uparrow$	R _F ^p $\uparrow$	R _F ^p $\uparrow$	R _C ^p $\uparrow$	R _C ^p $\uparrow$	R _F ^s $\uparrow$	R _F ^s $\uparrow$	R _C ^s $\uparrow$	R _C ^s $\uparrow$	R _F ^a $\uparrow$	R _F ^a $\uparrow$	R _C ^a $\uparrow$	R _C ^a $\uparrow$
Best Individual	baseline	0.266	0.259	0.284	0.294	-0.052	-0.038	-0.005	-0.128	0.027	0.010	0.050	-0.078	-0.016	0.020	-0.009	0.007	0.045	-0.038	0.043	-0.039
Simple Averaging	matched NAIVE	0.220	0.344	0.243	0.395	-0.037	-0.036	0.004	-0.137	0.001	-0.005	0.031	-0.094	-0.012	0.011	-0.004	-0.002	0.042	-0.042	0.043	-0.042
Simple Averaging	IND	0.209	0.394	0.236	0.447	-0.014	-0.015	0.035	-0.110	0.005	0.002	0.033	-0.089	-0.017	-0.009	-0.011	-0.024	0.048	-0.055	0.046	-0.053
Borda	matched NAIVE	0.232	0.353	0.256	0.408	-0.042	0.000	-0.001	-0.089	-0.020	-0.007	0.010	-0.096	-0.008	0.007	-0.006	-0.003	0.043	-0.041	0.035	-0.042
Borda	IND	0.226	0.379	0.247	0.430	-0.027	0.051	0.021	-0.043	-0.018	0.041	0.013	-0.055	-0.006	0.027	-0.002	0.012	0.047	-0.034	0.047	-0.033
Kemeny–Young	matched NAIVE	0.244	0.360	0.263	0.415	-0.041	0.011	0.003	-0.086	-0.002	0.012	0.031	-0.079	-0.009	0.017	-0.002	0.007	0.044	-0.035	0.044	-0.036
Kemeny–Young	IND	0.241	0.377	0.258	0.431	-0.029	0.038	0.031	-0.048	-0.012	0.046	0.019	-0.052	-0.004	0.024	0.008	0.016	0.043	-0.028	0.043	-0.027
RRF	matched NAIVE	0.245	0.342	0.265	0.395	-0.036	0.002	0.006	-0.093	-0.008	0.000	0.023	-0.091	-0.005	0.013	0.003	0.000	0.047	-0.037	0.045	-0.040
RRF	IND	0.252	0.346	0.271	0.406	-0.030	0.047	0.029	-0.034	0.004	0.032	0.028	-0.059	-0.015	0.033	0.002	0.028	0.040	-0.030	0.040	-0.025
Schulze	matched NAIVE	0.244	0.358	0.267	0.412	-0.049	0.008	-0.005	-0.088	-0.014	0.007	0.018	-0.082	-0.014	0.017	-0.008	0.004	0.038	-0.032	0.036	-0.034
Schulze	IND	0.227	0.373	0.247	0.426	-0.011	0.045	0.038	-0.048	-0.012	0.035	0.020	-0.057	0.007	0.028	0.018	0.016	0.048	-0.041	0.051	-0.040

Table 3: NOISE-S results for OctMNIST with ViT-B/16 (Full-matrix protocol), using the Spearman-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Simple Averaging	6	0.260	0.211	0.289	0.246	0.001	-0.071	0.055	-0.145	0.071	-0.029	0.091	-0.093	0.011	0.001	0.022	-0.006	0.026	-0.110	0.025	-0.108
Borda	6	0.270	0.219	0.298	0.249	-0.013	-0.075	0.025	-0.166	0.046	-0.035	0.062	-0.113	-0.011	0.016	-0.006	0.001	0.004	-0.064	0.010	-0.070
Kemeny–Young	6	0.263	0.179	0.303	0.211	0.019	-0.061	0.061	-0.134	0.076	-0.019	0.079	-0.090	0.015	-0.002	0.016	-0.010	0.007	-0.079	0.005	-0.078
RRF	6	0.274	0.192	0.303	0.223	-0.006	-0.067	0.052	-0.141	0.066	-0.022	0.077	-0.084	-0.007	0.012	0.004	0.004	0.008	-0.074	0.015	-0.074
Schulze	6	0.268	0.192	0.300	0.229	0.011	-0.067	0.054	-0.136	0.062	-0.011	0.078	-0.077	0.006	-0.004	0.016	-0.007	0.015	-0.076	0.011	-0.076

Table 4: NOISE-K results for OctMNIST with ViT-B/16 (Full-matrix protocol), using the Kendall-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Simple Averaging	3	0.267	0.200	0.312	0.227	0.041	-0.023	0.066	-0.101	0.076	-0.003	0.077	-0.088	-0.001	0.005	0.006	-0.004	-0.020	-0.063	-0.026	-0.064
Borda	3	0.270	0.194	0.313	0.222	0.027	-0.028	0.053	-0.108	0.058	0.004	0.063	-0.080	-0.009	0.002	-0.005	-0.007	-0.003	-0.064	-0.004	-0.064
Kemeny–Young	3	0.260	0.202	0.310	0.231	0.047	-0.029	0.068	-0.103	0.073	0.003	0.071	-0.075	0.007	0.011	0.003	0.004	-0.011	-0.062	-0.018	-0.063
RRF	3	0.270	0.191	0.316	0.216	0.025	-0.046	0.046	-0.127	0.068	0.000	0.067	-0.089	-0.006	0.001	-0.006	-0.009	-0.014	-0.050	-0.014	-0.051
Schulze	3	0.266	0.200	0.312	0.231	0.036	-0.020	0.059	-0.095	0.062	0.012	0.065	-0.066	0.003	0.003	0.003	-0.004	-0.011	-0.055	-0.011	-0.052